

Leaderboard & Matchmaking Use Cases

See Personal Stats

Iteration: 1, last modification: March 7, 2025

Adding personal stats to Matchmaking & Leaderboard would result in low cohesion. It would be preferable to have the authentication & profile team handle it.

Description	Passing the player's data to profile for user to view the personal statistics
Primary Actor	Game Statistics Class
Goal in Context	To pass along data from the database to profile and UI to display the person's data
Preconditions	1. The player must be logged in and the user ID exists in the database 2. There must be a request from the player via the UI 3. The data must be converted from SQL to Java
Trigger	The person's data will be converted from Queries to Java then profiles which will be displayed on the UI
Scenario	1. The player clicks on the profile button 2. It will send a request to Game Statistics Class which will get data from the database and converting it to Java 3. It will send the data to profile 4. Profile will display the data on UI
Exceptions	1. Data not found - Personal data was not found in the database
Priority	High. This is a core system requirement
When available	Third Iteration
Frequency of Use	Once a day
Channel to Actor	Interaction with the user profile menu UI
Secondary Actors	Database
Channel to Secondary Actors	Queries to the database
Open Issues	1. How to get data from the database team, how will we communicate with the database team 2. What type of data should we pass to user profile

a high priority system requirement would be better moved to second iteration to give time for testing. I would also reconsider the priority rating, as the feature is more important to player experience rather than core system requirement

Match Making

Iteration: 1, last modification: March 7, 2025

Description	To put the player into a queue to find a game by changing the player's status as interested in the database		
Primary Actor	Player		
Goal in Context	To put the player in a match's queue and change their status in the database as interested		
Preconditions	1. The player must be logged in with a valid ID 2. Internet connection to the database must be made 3. The player must select a game that they are interested in playing		
Trigger	Update database and change the player status in a game to be interested		
Scenario	1. The player clicks on a game they want to play 2. The player clicks on the find a match button 3. That will send an update to the match making database class 4. That will send a Query to the database		
Exceptions	1. Player ID not found - The player ID does not match the player ID 2. No game selected - The player did not select a game 3. Suitable opponent is not found - No player with similar elo		
Priority	High. This is an important function since this would be the easiest way to enter a match		
When available	Third Iteration	The availability for an important feature required to play the games would be better accomodated by implementing in the first iteration. This leaves enough room for testing and integration.	
Frequency of Use	Several times a day		
Channel to Actors	Interaction with game selection menu UI		
Secondary Actors	Database		
Channel to Secondary Actors	Queries to the database		
Open Issues	How will we update and find a suitable match using the database		

Challenge A User

Iteration: 1, last modification: March 7, 2025

Description	Interaction with a friend's invitation to start a game
Primary Actor	Player
Goal in Context	To accept or deny a request for a game invite
Preconditions	1. The player must be logged in with a valid ID 2. A request must be sent to the player
Trigger	Either abort the Match creation or create a Match variable that will be passed to game logic server code
Scenario	1. The player sends out an invitation to their friend 2. The player's friends will either accept or deny the request through the UI 3. .That will either trigger the creation of Match in the server or Not.
Exceptions	1. Player ID not found - Either the player or player's friend ID is not found 2. Match no longer exist - If the match invites times out
Priority	Medium. This is an important function but automatically finding a match is more important
When available	Third iteration
Frequency of Use	Several times a day
Channel to Actors	Interaction with game selection menu
Secondary Actors	Player's friends
Channel to Secondary Actors	Via networking to send the invite and receive the answer
Open Issues	1. How would we go about sending the invites over the network? 2. How long should the invite last before canceling the invite 3. How would we put both people into a queue so the system knows when to start the game with those two people

Issue no. 3 would have to be handled by the matchmaking system as part of core functionality. I suggest defining the matchmaking system and its functionality more explicitly, as part of planning stage

Start a Party Match

Iteration: 2, last modification: March 7, 2025

Description	Starting a match for all players in the party
Primary Actor	Players in the party
Goal in Context	To create a Match with the players in the party as its variable
Preconditions	1. Both players must be logged in with a valid player's ID 2. Both players must be in the same party and the party size must be 2 3. A game must be selected prior to starting a match
Trigger	A creation of Match in MatchMaking Class with both player's ID as its parameter
Scenario	1. Both players are in the same party 2. The party's leader will select a game to start 3. The party's leader the click the start button which will get both players into a game
Exceptions	1. Invalid player's ID - If one or both players have a non- matching ID with the database 2. Invalid party's size - If the party size is not 2 3. Game selection missing - If a game is not selected
Priority	Medium. This is an important function but automatically finding a match is more important
When available	Third Iteration
Frequency of Use	Several times a day
Channel to Actors	Interaction with game selection menu
Secondary Actors	None
Channel to Secondary Actors	N/A
Open Issues	1. We are not too sure on how we will check the party size system

View Personal Leaderboard

Iteration: 3, last modification: March 7, 2025

Description	Make a list of 50 players that are above and below the players current rank and pass the list to GUI	
Primary Actor	Player	
Goal in Context	A player wants to view where they are in relation to player at the same skill level	
Preconditions	1. The player is logged in with a valid player's ID 2. The player navigated to their profile	
Trigger	Make a sorted list of 50 players that are directly ranked above and below them	I suggest making the trigger an action or a button to view player's personal leaderboard.
Scenario	1. The player logs in 2. The player navigated to their profile 3. The player wants to check their standing with players of similar elo	
Exceptions	1. Invalid player's ID - If the player's ID does not match the database 2. Sever Interruption - If the connection with the database is interrupted	

Priority	Low. This is a quality-of-life feature that will encourage players, and help them to be more engaging with the game		
When available	Iteration 3	Considering the priority and iterations assigned to the other use cases, it would be preferable to move the availability to a later iteration.	
Frequency of Use	However often the players want to know their rank		
Channel to Actors	Interaction with game UI		
Secondary Actors	Database		
Channel to Secondary Actors	Database's queries to get player's data		
Open Issues	1. Have not picked a data type to pass to GUI		

View Friend's List

Iteration: 3, last modification: March 7, 2025

Description	Returns a sorted list of all the other players that have been friended and sorted them by elo.	
Primary Actor	Player	
Goal in Context	To view the player elo in their friend's list	
Preconditions	1. The player opens the game 2. The game is connection to the internet 3. The player is logged in 4. The game is connected to the internet 5. The player navigates to the friend's list	
Trigger	Gives an up-to-date list of all their friends sorted by their elo	This description better fits the post-condition, in comparison to the trigger
Scenario	1. The player logs in 2. The player navigates to the friend's list 3. The player wants to check where they stand in elo ranking to their friends	
Exceptions	1. Invalid player's ID - If the player's ID does not match the database 2. Sever Interruption - If the connection with the database is interrupted	

Priority	Low. This is a quality-of-life feature that boost the competitiveness of the player and encourage interactions with the game
When available	Iteration 3
Frequency of Use	Dependent on how serious the player is about the game, ranging from after every match to once per day.
Channel to Actors	Interaction with game UI.
Secondary Actors	None
Channel to Secondary Actors	N/A
Open Issues	Coordinating with GUI

View Top Players Leaderboard

Iteration: 2, last modification: March 7, 2025

Description	Provides a list of global player base sorted by elo	
Primary Actor	Player	
Goal in Context	To have a list of the top 500 players sorted by elo	
Preconditions	1. The player opens the game 2. The game is connection to the internet 3. The player logs in 5. The player navigate to the leaderboard	
Trigger	Show the leaderboard with the top 500 players	
Scenario	1. The player logs in 2. The player selects the leaderboard button 3. The game will show a list of the top 500 players that the player can scroll through	
Exceptions	1. Invalid player's ID - If the player's ID does not match the database 2. Sever Interruption - If the connection with the database is interrupted	
Priority	Low. This is a quality-of-life feature that can be implemented later	
When available	Iteration 3	I suggest pushing back the availability iteration, as high, medium and low priority features are all listed in the third iteration for all features. This would not leave enough time for proper testing and integration.
Frequency of Use	Low, once per day	

Channel to Actors	Interaction with game UI.	
Secondary Actors	None	
Channel to Secondary Actors	N/A	
Open Issues	Coordinating with GUI	